FemboyEngine API Reference

This is the canonical API index. There is**one API document**, not one per OS.

Platform differences are inline notes. Build, deployment and unsupported-target

details live in PLATFORMS.md.

Platform tags

Tag	Meaning
Desktop	Windows, Linux, macOS
Windows	Implemented only on Windows
Unix	Linux and macOS
Desktop backend	Requires the SFML desktop window/audio backend
Mobile: no	Not available until a separate platform layer exists
Backend	Behaviour belongs to the renderer/audio backend, not the API shape

If an entire module is unavailable on mobile, it is marked once here and once in

PLATFORMS.md; individual methods are not duplicated per OS.

Core

Headers: FemboyEngine/Core.hpp, Engine.hpp, FemObject.hpp, Result.hpp,

Time.hpp, Config.hpp.

- Engine — application lifetime, window, fixed-step loop, renderer.
- Application — layer stack and composition root.
- FemObject / FemClass — engine object model and reflection registration.
- ProfileScope, Profiler — frame and system timing.
- Result<T> — explicit success/error without exceptions.
- Time, DateTime — monotonic and wall-clock time.

Platform: Desktop. Mobile needs a platform loop and lifecycle bridge.

Console

Headers: FemboyEngine/Console.hpp, DebugConsole.hpp.

- Console — in-game developer console, command registry, history and overlay.

Renders through DrawList; available on all desktop platforms.

- DebugConsole — optional OS console for early diagnostics.

- Windows: AllocConsole + stdout/stderr/stdin redirect.
- Linux/macOS: no-op; a terminal launch already has a console.
- Mobile: no — use platform logs.
- Toggle: F1 or .
- --console / --dev-console requests the OS console.

Math

Header: FemMath/Math.hpp (separate library, no ECS/render dependency).

- Vec2, Vec2i, Color, Rect.
- Pi, Tau, DegreesToRadians, RadiansToDegrees.
- Length, Normalized, Dot, Cross, Rotated, FromAngle, AngleOf.
- Distance, DistanceToSegment, RectContains.
- Lerp, WithAlpha, Clamp01.

Platform: Desktop and future mobile; no OS-specific behaviour.

World and scenes

Headers: FemboyEngine/World.hpp, ECS.hpp, Scene.hpp, Prefab.hpp,

ClassDefaults.hpp, Mirror.hpp.

- World — entity lifecycle: Spawn, SpawnWith, Destroy, Clear, Valid,

Count, View, Get, TryGet, Emplace, Has.

- Scene — JSON scene load/save and additive application.
- PrefabLibrary / Prefab — inheritance, overrides and nested children.
- ClassDefaults — project defaults between C++ defaults and prefab overrides.
- Mirror — reflected component fields and editor metadata.

Platform: all. File paths are handled by Vanity and differ by OS.

Rendering

Headers: FemboyEngine/Skirt.hpp (DrawList), IRenderer.hpp,

Camera2D.hpp, ViewProjection.hpp, Tilemap.hpp, Lighting.hpp,

PostFX.hpp, Display.hpp.

- DrawList — layered command list, stable fractional depth sort and batching.

Public name is DrawList; Skirt is a compatibility alias.

- IRenderer — backend contract. SFMLRenderer is the shipped desktop backend.
- Camera2D / ViewProjection — orthographic and affine isometric projection,

world/screen conversion.

- Tilemap::RenderSorted — diagonal depth ordering for isometric maps.
- DisplaySettings, PostFX — frame presentation and screen effects.

Platform:

- Desktop backend only today.
- Vulkan is a future backend, not a current build option.
- Mobile: no until EGL/Metal + platform renderer land.

Input, UI and accessibility

Headers: FemboyEngine/Input.hpp, UI.hpp, Accessibility.hpp.

- Input — actions, mapping contexts, modifiers, rebinding, save/load.
- PinkUI — JSON widget trees rendered through DrawList.
- Accessibility — colorblind/high-contrast adjustment, reduce motion/flash,

subtitles, TTS.

Platform:

- Desktop: keyboard/mouse/gamepad.
- Touch input is part of the future mobile platform layer.
- TTS: Windows SAPI; Unix logs unless the host supplies a callback.
- Subtitles and visual accessibility flags are platform-neutral.

Audio

Headers: FemboyEngine/Audio.hpp, AudioMixer.hpp, AudioOcclusion.hpp,

VoiceChat.hpp.

Platform:

- Desktop: SFML/miniaudio backend.
- Device names and availability differ by OS.
- Mobile: no until a platform audio backend is added.

Assets and files

Headers: FemboyEngine/Assets.hpp, Image.hpp, ResourcePack.hpp,

AssetPipeline.hpp, Vanity.hpp.

- AssetManager — textures, fonts, sounds, music, async decoding.
- AssetDatabase / AssetCooker — manifest validation and cooking.
- Vanity — atomic reads/writes, path joins, upward search, user data dir.

Platform:

- Desktop paths: %LOCALAPPDATA%, ~/Library/Application Support,

$XDG_DATA_HOME.

- Mobile needs AAssetManager / bundle resource bridges.

Networking and physics

Headers: FemboyEngine/Networking.hpp, Hosting.hpp, Replication.hpp,

Rollback.hpp, Physics.hpp, PhysicsSystem.hpp.

Platform: Desktop (ENet + Box2D). No API shape change on other platforms, but

mobile packaging and lifecycle handling are separate work.

Scripting and tools

Headers: FemboyEngine/FemboyScript.hpp, VisualScript.hpp, Reflection.hpp,

ProjectGen.hpp.

- FemboyScript — text scripting VM.
- VisualScriptGraph — node graph compiled to FemboyScript.
- FemStructRegistry / FemEnumRegistry — reflection metadata.
- ProjectGenerator — editor/tooling only.

Platform: scripting and reflection are neutral; the editor and project generator

are desktop tooling.

Rules for API changes

- Do not duplicate an API page per OS.
- Add a Platform: block only when behaviour differs.
- Mark a whole module once if it is unavailable on a platform.
- Keep signatures, JSON shapes and serialized enum values stable.
- A new platform backend implements IRenderer/input/audio seams; it does not

fork the public API.